European Journal of Neuroscience, Vol. 20, pp. 264–272, 2004 © Federation of European Neuroscience Societies

Microstimulation of V1 delays the execution of visually guided saccades

Edward J. Tehovnik, Warren M. Slocum and Peter H. Schiller
Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, E25-634, Cambridge, MA 02139

Keywords: current spread, electrical stimulation, occipital cortex, phosphenes, rhesus monkeys, visual prosthesis

Abstract

Electrical stimulation delivered to V1 concurrently with the presentation of a visual target interferes with both the selection and the detection of targets positioned in the receptive field of the stimulated neurons. In the present study, we examined the temporal course of this effect by delivering electrical stimulation to V1 of rhesus monkeys at various times before the appearance of a visual target. Each trial was initiated by the appearance of a fixation spot that, once acquired, was followed by the presentation of a visual target in the receptive field of the stimulated neurons. A monkey was reward after making a saccadic eye movement to the target. A delay in saccade generation was obtained when stimulation was delivered while an animal maintained fixation on the fixation spot. No delay occurred when the visual target was placed outside the receptive field of the stimulated neurons. The best parameters for inducing the saccadic delay were: (i) anode-first pulses (as opposed to cathode-first pulses) and (ii) train durations greater than 40 ms and frequencies greater than 100 Hz. The lowest current threshold for producing a saccadic delay occurred at 1.5 mm below the top of superficial V1. The chronaxies of the directly stimulated elements mediating the delay ranged from 0.13 to 0.24 ms. These values overlap with those that have been described for phosphene induction in human V1. We discuss how the elements mediating the saccadic delay might interrupt a visual signal as it passes along the geniculostriate pathway.

Introduction

Electrical stimulation of V1 in humans has been shown to produce a visual percept, called a phosphene (Brindley & Lewin, 1968a,b; Dobelle & Mladejovsky, 1974; Schmidt *et al*., 1996). These findings have led to the suggestion that stimulation of V1 in monkeys using arrays of electrodes could serve as a model for a visual prosthetic in the blind (Troyk *et al*., 2003).

Recent work has uncovered several effects of electrical stimulation of V1 in rhesus monkeys. Electrical stimulation of V1 typically elicits a saccadic eye movement that shifts the centre of gaze to the receptive field of the stimulated neurons (Tehovnik *et al*., 2003a). The effectiveness with which saccades are evoked depends on what layers are being stimulated and on when the stimulation is delivered. Saccade generation occurs most readily when the deepest layers of V1 are stimulated (Tehovnik *et al*., 2003a). Stimulation that is delivered after the dousing of the fixation spot is best at producing such saccades (Tehovnik *et al*., 2003b). When stimulation is delivered while the fixation spot is on, currents in the milliampere range fail to evoke saccades throughout V1.

When electrical stimulation is delivered concurrently with the presentation of a visual target placed in the receptive field of the stimulated neurons, saccades generated toward the receptive-field target can either be facilitated or interfered with depending on the cortical layers stimulated (Schiller & Tehovnik, 2001; Tehovnik *et al*., 2002a,b; Tehovnik & Slocum, 2003a). In the lower layers of V1 facilitation is most commonly obtained; the stimulation increases the probability of saccades and decreases the latency of saccades made to the receptive-field target. By contrast, in the upper layers stimulation generally produces interference; the probability with which a saccade is generated decreases and the latency of saccade initiation increases to the receptive-field target. These effects are spatially local; electrical stimulation is ineffective when the visual target is placed at locations outside the receptive field of the stimulated neurons.

In studies carried out so far examining the interference effect, electrical stimulation was delivered concurrently with the presentation of the visual target. In the current study, we examined the temporal course of this effect by delivering electrical stimulation to V1 of rhesus monkeys at various times before the appearance of the visual target. We found that stimulating V1 while an animal was fixating a spot of light systematically delayed the execution of the subsequent visually guided saccade. Our study had four goals: (i) to determine the optimal stimulation parameters for the production of the saccadic delay; (ii) to determine whether the delay effect is confined to the receptive-field location of the stimulated neurons; (iii) to determine the cortical depth at which the delay effect is most pronounced and (iv) to determine the excitability properties of the directly stimulated elements mediating the delay. We relate this work to studies on the induction of visual phosphenes by electrical stimulation of human V1.

Materials and methods

Subjects

Two adult rhesus monkeys (*Macaca mulatta*) M and D were used. Throughout this study food was freely available. A monkey's access to

Correspondence: Dr Edward J. Tehovnik, as above.
E-mail tehovnik@mit.edu

Received 3 February 2004, revised 2 April 2004, accepted 5 May 2004

doi:10.1111/j.1460-9568.2004.03480.x

water was limited before each day of experimental testing. After testing, the monkeys were allowed to drink until sated before being returned to the vivarium. Monkeys were provided for in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

Surgery

Monkeys were anaesthetized intravenously with pentobarbital (20 mg/kg) and prepared for aseptic surgery. A scleral search coil was implanted (Judge *et al*., 1980), and a stainless-steel post, used to restrain the head, was secured to the skull with titanium screws and acrylic cement. Subsequently, a recording chamber was implanted over the right V1.

Behavioural tasks

A monkey, with head fixed in position, faced a computer monitor (Sony Multiscan E210) positioned 57 cm away in a dimly lit room. The animal was trained to first fixate a central spot and to then make a saccade to a visual target that appeared at various locations on the monitor screen. This task had two forms. In the first, the fixation spot remained on for 2–4 s and the animal had to maintain fixation until the spot was terminated and a target appeared. While the animal maintained fixation, a bar was swept across a restricted region of the monitor to map the receptive field of the neurons recorded from V1. After each sweep, the target appeared and the animal was rewarded with a drop of apple juice after a saccade was made to the target. The orientation and size of the bar as well as the rate and distance at which the bar moved were varied systematically until the location of the receptive field was accurately determined (Fig. 1A, right).

For the second task, a monkey was required to fixate a spot of light (0.1 degree in diameter at 100% contrast) for 600 ms after which a visual target (0.2 degrees in diameter at 100% contrast) was presented in one of several locations for 500 ms. One of these locations was in the receptive field of the neurons under study. Monkeys received a juice reward for generating a saccade to the target within 500 ms following target onset (Fig. 1Ba). During fixation, an animal was required to keep his eyes within a 0.5 by 0.5 degree window; otherwise the trial was aborted. During experimental sessions, electrical stimulation was delivered to V1 on 50% of trials in a random order (Fig. 1Bb). The parameters of stimulation and the time of its delivery relative to the onset of the fixation spot and its termination were systematically varied. The latency with which saccades were generated constituted the prime data for the study.

Estimating electrode depth

For all electrode penetrations made into cortex (Fig. 1A, left), a standard method was used to deduce the electrode depth with respect to the top of superficial V1 (Tehovnik *et al*., 2002a, 2003a; Tehovnik & Slocum, 2003b). Once single units were encountered, the receptive field was mapped and the electrode was adjusted in depth until the unit activity was just detectable at the top of V1. This point marked the initial depth at the top of V1. At the end of a test session, the depth at the top of V1 was remeasured by noting the point at which unit activity to visual stimuli became barely detectable as the electrode was withdrawn. The estimated depth at the top of V1 was taken as the average of the initial and final depth estimates.

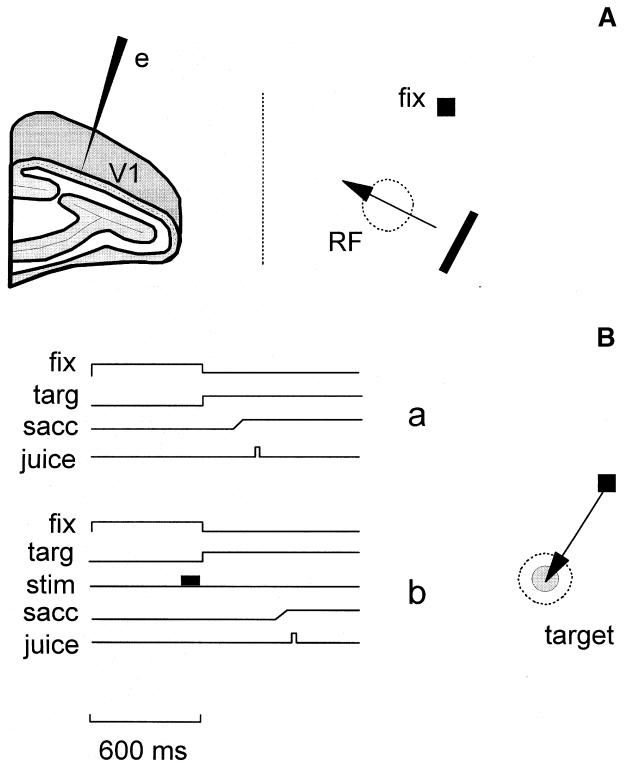

FIG. 1. Methods. (A, left) All electrodes (e) were lowered into V1 such that they penetrated V1 perpendicular to the cortical surface. (A, right) Receptive fields (RF) were mapped by moving a bar of light at different orientations across the visual field as the monkey remained fixated (fix). (B) For all trials, an animal fixated a spot for 600 ms (fix) otherwise the trial was aborted and no juice reward delivered. Two types of trials were interleaved to study stimulation-evoked interference. (a) Following fixation, a target (targ) was presented in the receptive-field location of the stimulated neurons following the extinction of the fixation spot (also see illustration to right). The animal was required to generate a saccade (sacc) to the target within 500 ms to obtain a juice reward (juice). (b) Following fixation, a target (targ) was presented in the receptive field of the stimulated neurons following the extinction of the fixation spot. Before the end of the fixation period, a train of electrical stimulation (stim) was delivered to the neurons. The animal was required to generate a saccade (sacc) to the target within 500 ms to obtain a juice reward (juice).

Data collection and analysis

A PDP−11/73 computer controlled the presentation of visual stimuli, the delivery of electrical stimulation, the collection and display of eye position (sampled at 200 Hz), and the delivery of juice.

Electrical stimulation

Constant-current charge-balanced biphasic pulses were delivered to the brain via a monopolar glass-coated platinum-iridium electrode [at 1–2 MΩ tested at 1 kHz] using a Grass S88 stimulator attached to a pair of constant-current stimulus isolation units (Grass PSIU6B, Quincy, MA, USA). For each biphasic pulse, an anodal and cathodal pulse followed in immediate succession to produce anode-first pulses. Cathode-first pulses were used for some experiments. Both pulses of a pair had the same amplitude and duration. Current was measured by the voltage drop across a 1000 Ω resistor that was in series with the return lead of the stimulator. The current was monitored continuously using an oscilloscope and was read as the amplitude of one pulse (anodal or cathodal) of a biphasic pair.

Electrodes were introduced perpendicular to the cortical surface with a hydraulic microdrive (Fig. 1A, left). The action potentials were

amplified (Bak A−1B, Germantown, MD, USA) and filtered (Krohn-Hite 3750, Horsham, PA, USA). The stimulation parameters used were as follows: current ranged from 2 to 200 μA; the pulse duration ranged from 0.05 to 0.7 ms, the pulse frequency ranged from 10 to 400 Hz, and train duration ranged from 20 to 200 ms.

Results

Stimulation sites

For monkey M, the penetrations contained cells with receptive-field centres varying from 2.8 to 4.6 degrees of eccentricity and from 225 to 250 degrees of meridian. For monkey D, the penetrations yielded cells with receptive-field centres varying from 2.8 to 3.1 degrees of eccentricity and from 241 to 250 degrees of meridian. A total of 24 and seven penetration sites were studied in monkeys M and D, respectively. For other details regarding the V1 stimulation sites see Tehovnik *et al*. (2002a).

Parameters of electrical stimulation

Electrical stimulation during the fixation period according to the scheme of Fig. 1B was found to delay the execution of visually guided saccades generated to a visual target positioned in the receptive field of the stimulated neurons. Several parameters of stimulation were found to affect the robustness of this effect. First, anode-first pulses were more effective than cathode-first pulses in increasing the latency of visually guided saccades (Fig. 2A). Second, increases in the frequency of pulses delivered with the stimulation train increased the latency of visually guided saccades such that beyond 100 Hz there was little increase (Fig. 2B). Third, the time at which electrical stimulation was delivered played a central role in the increase of saccadic latencies. To produce an increase in latency, the train of stimulation had to be delivered at the end of the fixation period (Fig. 3A). Fourth, increases in train duration increased the latency of visually guided saccades such that beyond 40 ms there was little further increase (Fig. 3B). Finally, increases in current increased the latency of visually guided saccades (Fig. 4). Based on these results, the following parameters of stimulation were used for the remaining experiments: anode-first pulses were used with train duration, pulse duration, and frequency fixed at 100 ms, 0.2 ms, and 200 Hz, respectively. All stimulation trains commenced 100 ms before the end of the fixation period using the scheme of Fig. 1B. Currents from 2 to 200 μA were used for the remaining experiments.

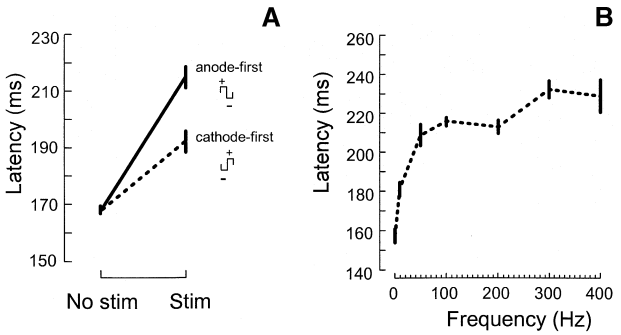

FIG. 2. Effect of pulse-pair configuration and pulse frequency on saccadic latency. (A) The latency of visually guided saccades to the receptive-field target is plotted as a function of whether or not stimulation (Stim vs. No stim) is delivered during the fixation period and whether anode-first or cathode-first pulses are delivered. Each value is based on 20 trials. Standard errors are shown. The latency was greater following the delivery of anode-first pulses as compared to the delivery of cathode-first pulses ($t_{19} = 5.07$, $P < 0.01$). Also the latency was prolonged for stimulation trials as compared to non-stimulation trials for both the anode-first and cathode-first conditions (no stimulation vs. anodal stimulation: $t_{19} = 12.5$, $P < 0.01$; no stimulation vs. cathodal stimulation: $t_{19} = 9.0$, $P < 0.01$). The parameters of stimulation were as follows: current, train duration, pulse duration, and frequency were fixed at 50 μA, 100 ms, 0.2 ms, and 200 Hz, respectively. Stimulation started 500 ms into the fixation period. The depth of stimulation was at 1.75 mm below the cortical surface. The receptive-field location of the stimulated units was at 234 degree of meridian and at 3.4 degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. For other details see Materials and methods and Fig. 1B. Data from monkey M. (B) The latency of visually guided saccades to the receptive-field target is plotted as a function of frequency of stimulation. Each value is based on 20 trials. Standard errors are shown. The parameters of stimulation were as follows: anode-first pulses were used and the current, train duration, and pulse duration were fixed at 200 μA, 100 ms, and 0.2 ms, respectively. Stimulation started 500 ms into the fixation period. The depth of stimulation was at 1.5 mm below the cortical surface. The receptive-field location of the stimulated units was at 250 degree of meridian and at 3.6 degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. For other details see Materials and methods and Fig. 1B. Data from monkey M.

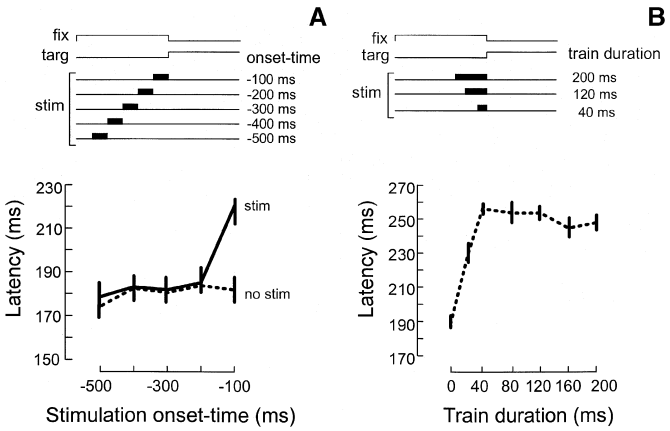

FIG. 3. Effect of stimulation-onset time and train duration on saccadic latency. (A) The latency of visually guided saccades to the receptive-field target is plotted as a function of stimulation-onset time with respect to the termination of the fixation spot. Data from non-stimulation (no stim) and stimulation trials (stim) are shown. Each value is based on 20 trials. Standard errors are shown. The latency was greater for stimulation trials than for non-stimulation trials when the stimulation onset-time was −100 ms ($t_{19} = 4.9$, $P < 0.01$). The parameters of stimulation were as follows: anode-first pulses were used and current, train duration, pulse duration, and frequency were fixed at 100 μA, 100 ms, 0.2 ms, and 200 Hz, respectively. The depth of stimulation was at 0.7 mm below the cortical surface. The receptive-field location of the stimulated units was at 245 degree of meridian and at 3.7 degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. For other details see Materials and methods and Fig. 1B. Data from monkey M. Details pertaining to the onset-time of the delivery of electrical stimulation (stim) with respect to the end of the fixation period (fix) are shown above. Target time (targ) is indicated. (B) The latency of visually guided saccades to the receptive-field target is plotted as a function of the duration of the stimulation train. Each value is based on 20 trials. Standard errors are shown. The parameters of stimulation were as follows: anode-first pulses were used and the current, pulse duration, and frequency were fixed at 100 μA, 0.2 ms, and 200 Hz, respectively. Stimulation offset-time occurred at the end of the fixation period. The depth of stimulation was at 1.75 mm below the cortical surface. The receptive-field location of the stimulated units was at 234 degree of meridian and at 3.4 degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. For other details see Materials and methods and Fig. 1B. Data from monkey M.

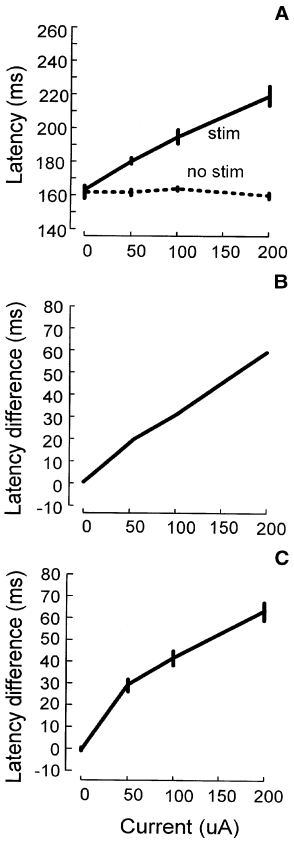

FIG. 4. Effect of stimulation current on saccadic latency. (A) The latency of visually guided saccades to the receptive-field target is plotted as a function of stimulation current. Data from non-stimulation (no stim) and stimulation trials (stim) are shown. Each value is based on 20 trials. Standard errors are shown. The parameters of stimulation were as follows: anode-first pulses were used and train duration, pulse duration, and frequency were fixed at 100 ms, 0.2 ms, and 200 Hz, respectively. The depth of stimulation was at 0.75 mm below the cortical surface. The receptive-field location of the stimulated units was at 235 degree of meridian and at 3.5 degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. For other details see Materials and methods and Fig. 1B. Data from monkey M. (B) Latency difference between stimulation and non-stimulation trials for the generation of visually guided saccades to the receptive-field target is plotted as a function of stimulation current. Data from A above. (C) Average latency difference between stimulation and non-stimulation trials for the generation of visually guided saccades to the receptive-field target is plotted as a function of stimulation current for 35 stimulation sites, 19 sites from monkey M and 16 sites from monkey D. These sites were located between 0 and 2.25 mm below the cortical surface. Standard errors are shown. The stimulation parameters used are described in A.

Location of the visual target

The location of the visual target with respect to the receptive-field centre of the stimulated neurons had an effect on the latency of visually guided saccades during stimulation trials. The visual target used for this experiment was a bright circular spot of light at 100% contrast and at 0.2 degrees in diameter. The maximal increase in saccadic latency due to stimulation with a 100 μA current was observed when the target was positioned at the centre of the receptive field of the stimulated neurons (Fig. 5A and B). The magnitude of the latency increase dropped systematically as the distance between the target and receptive-field centre was increased. For target eccentricities beyond 0.5 degrees from the receptive-field centre, the stimulation became ineffective (Fig. 5A). Also for target meridian values outside of the receptive-field of the stimulated neurons, the stimulation became less effective (i.e. at 10 degrees of meridian on either side of the receptive field; Fig. 5B).

The above effect was studied using three levels of current. The latency difference for stimulation trials as compared to non-stimulation trials was computed for 25, 50, and 100 μA currents while target position was varied. The greatest latency difference was observed for all current conditions when the target was situated at the receptive-field centre of the stimulated neurons (Fig. 5C and D). The magnitude of this effect varied positively with current intensity.

Size of the visual target

The size of the visual target at the receptive-field centre of the stimulated neurons had an effect on the latency of visually guided saccades during stimulation trials. The visual target used for this experiment was a bright circular spot of light at 100% contrast whose target diameter was varied from 0.1 to 0.6 degrees. The maximal increase in saccadic latency due to stimulation with a 100 μA current was observed for the smallest target size, i.e. at 0.1 degrees in diameter (Fig. 6A). The magnitude of the latency increase dropped systematically as target size was increased.

The above effect was studied using three levels of current. The latency difference for stimulation trials as compared to non-stimulation trials was computed for 25, 50, and 100 μA currents while target size was varied. The greatest latency difference was observed for all current conditions when the target was the smallest, i.e. at 0.1 degrees in diameter (Fig. 6B). The magnitude of this effect varied positively with current intensity.

Current threshold to induce a saccadic delay as a function of cortical depth

The current to induce a 20-ms saccadic delay was determined as a function of cortical depth (Fig. 7). The 20-ms threshold criterion is based on data shown in Fig. 4C. Here the average latency difference of 20 ms is greater than three standard deviations of that observed when comparing non-stimulation and dummy stimulation trials (standard deviation = 4.3 ms, $n = 35$). The 20-ms threshold criterion was selected so that we could use currents below 100 μA to better reveal the laminar organization of V1 in regards to the stimulation-induced saccadic delay. Currents of 100 μA were not sufficient to evoke a 20 ms delay when the electrode was positioned between 0 and 0.5 mm below the cortical surface. Beyond 0.5 mm, the current threshold dropped, reaching a minimum of 26 μA, on average, at 1.5 mm below the cortical surface.

Chronaxie estimates of elements mediating the saccadic delay

In Fig. 8A, the current necessary to induce a 20-ms saccadic delay is plotted as a function of pulse duration for ten different sites. For all sites studied, the current threshold dropped with an increase in pulse duration reaching asymptote at a pulse duration of 0.6 ms. For the shortest pulse durations tested (i.e. 0.05 ms), the current threshold ranged from 34 to 174 μA and for the longest durations tested (i.e. 0.7 ms) the current threshold ranged from 4 to 50 μA.

To determine the excitability of the directly stimulated elements inducing the saccadic delay, the current-duration functions (Fig. 8A) were normalized such that the current threshold to evoke a 20-ms delay was set to one for a pulse duration of 0.7 ms and all other thresholds were expressed as a multiple of this threshold (Fig. 8B).

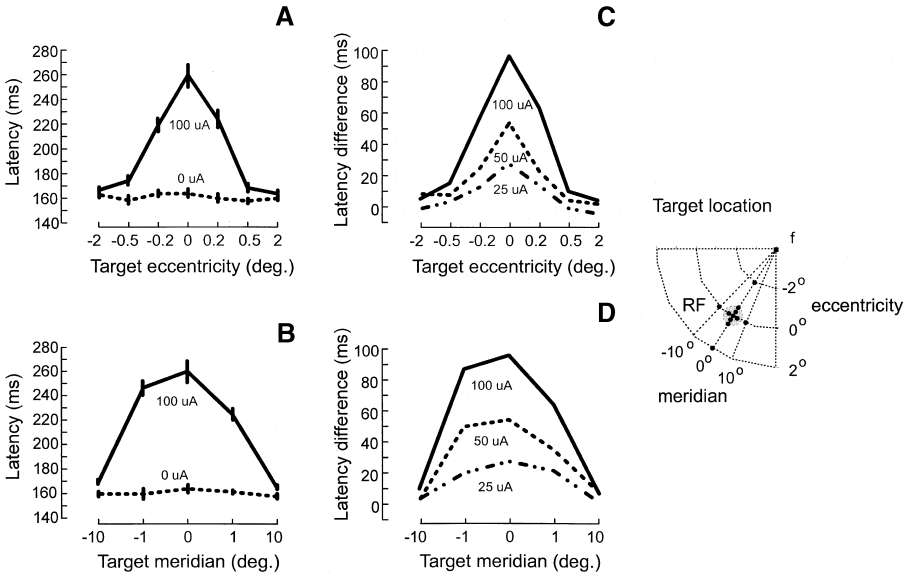

FIG. 5. Effect of stimulation on saccadic latency for different target positions at and outside of the receptive-field location of the stimulated neurons. (A) The latency of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive-field location of the stimulated neurons. A zero eccentricity along the *x*-axis indicates that the target and the receptive-field centre of the stimulated neurons were in register (see Target-location icon at the right of the figures). Negative values along the *x*-axis indicate target positions situated between the fixation position and receptive-field centre of the stimulated neurons. Positive values indicate target positions eccentric to the receptive-field centre of the stimulated neurons. The solid curve represents data from stimulation trials and the dashed curve represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The parameters of stimulation were as follows: anode-first pulses were used and current, train duration, pulse duration, and frequency were fixed at 100 μA, 100 ms, 0.2 ms, and 200 Hz, respectively. The depth of stimulation was 0.9 mm below the cortical surface. The receptive-field location of the stimulated units was at 237 degree of meridian and at four degrees of eccentricity. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. Target-location icon at right of figures: f represents the fixation location and RF represents the receptive-field location of the stimulated neurons. Data from monkey M. For other details see Materials and methods and Fig. 1B. (B) The latency of visually guided saccades to the target is plotted as a function of target meridian with respect to the receptive-field location of the stimulated neurons. A zero meridian value along the *x*-axis indicates that the target and the receptive-field centre of the stimulated neurons were in register (see Target-location icon at right of figures). Negative values along the *x*-axis indicate that the target positions had low meridian values, and positive values indicate that the target positions had high meridian values. The solid curve represents data from stimulation trials and the dashed curve represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. For other details see A. (C) The latency difference between stimulation and non-stimulation trials for the generation of visually guided saccades to the target is plotted as a function of target eccentricity with respect to the receptive-field location of the stimulated neurons for three levels of current: 25, 50, and 100 μA. Data for the 100-μA current level are from A. See A for other details. (D) The latency difference between stimulation and non-stimulation trials for the generation of visually guided saccades to the target is plotted as a function of target meridian with respect to the receptive-field location of the stimulated neurons for three levels of current: 25, 50, and 100 μA. Data for the 100-μA current level are from B. See B for other details.

Power functions were fitted for every data set pertaining to a site. For all data sets, the R^2 values were always greater than 0.8. The chronaxie value for a site can be determined as the pulse duration at which the power function crosses two units of threshold (Fig. 8B, dashed horizontal line). The chronaxie values ranged from 0.13 to 0.24 ms.

[A chronaxie is a measure of neuronal excitability such that axons have shorter chronaxies than cell bodies (axons, 0.03–7 ms; cell bodies, 7–31 ms; Ranck, 1975; Nowak & Bullier, 1998), and large, myelinated axons have shorter chronaxies than small, non-myelinated axons (large, 0.03–0.7 ms; small, > 1.0 ms; Ranck, 1975; Li & Bak, 1976; West & Wolstencroft, 1983).]

Current spread estimates

Current delivered to neural tissue through a microelectrode, spreads effectively proportional to the square root of the current divided by a current–distance constant (Tehovnik, 1996). For large pyramidal tract cells, this constant can range from 300 to 3000 μA/mm^2 with an average of approximately 1000 μA/mm^2 (Stoney *et al.*, 1968). These values were computed using a cathode current pulse having a duration of 0.2 ms. The current–distance constant reflects the excitability of a neural element 1 mm away from the electrode tip; an element having a constant of 1000 μA/mm^2 would require a 1000 μA current to be activated 1 mm away 50% of the time. The greater the current–distance constant the less the conduction velocity of an axonal element (Jankowska & Roberts, 1972; Roberts & Smith, 1973; Hentall *et al.*, 1984). Therefore the size of a neuron's axon, and whether it is myelinated or not, are related to the current–distance constant.

To estimate the spread of current in V1, we will assume a current–distance constant of 1000 μA/mm^2. This is a conservative estimate for V1 given that the neuronal elements in V1 of primates tend to be smaller and therefore less excitable than those of other parts of cerebral cortex (Cragg, 1967; Rockel *et al.*, 1980; O'Kusky & Colonnier, 1982; Fries, 1984; Peters, 1987). Indeed the conduction velocities of pyramidal neurons exiting V1 tend to be significantly lower than those of large pyramidal neurons (Finlay *et al.*, 1976; Macpherson *et al.*, 1982). Using the equation, $r = (I/K)^{1/2}$, where r is the distance of effective current spread from the electrode tip in mm, I is the current used in μA, and K is the current–distance constant in μA/mm^2, a 2–200 μA current (range used) delivered to V1 is estimated to activate elements directly within 5–447 μm, respectively, from the electrode tip.

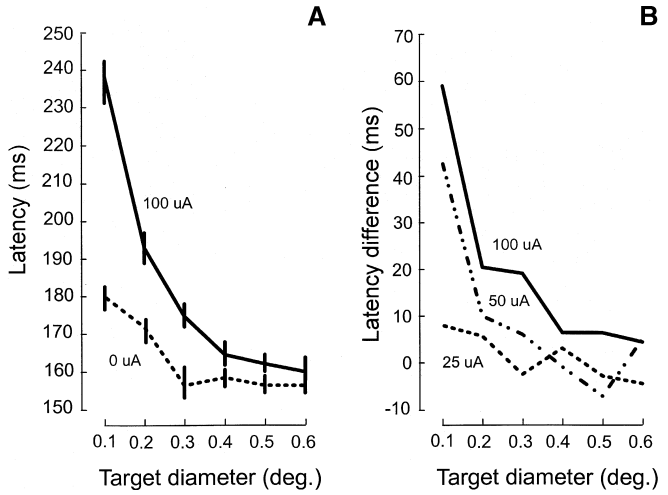

FIG. 6. Effect of stimulation on saccadic latency for different target sizes for a target centred on the receptive field of the stimulated neurons. (A) The latency of visually guided saccades to the target is plotted as a function of target diameter for stimulation and non-stimulation trials. The solid curve represents data from stimulation trials and the dashed curve represents data from non-stimulation trials. Each value is based on 20 trials. Standard errors are shown. The parameters of stimulation were as follows: anode-first pulses were used and current, train duration, pulse duration, and frequency were fixed at 100 μA, 100 ms, 0.2 ms, and 200 Hz, respectively. The depth of stimulation was 0.85 mm below the cortical surface. The receptive-field location of the stimulated units was at 245 degree of meridian and at 4.2 degrees of eccentricity. The target used was brighter than background at 100% contrast. Data from monkey M. For other details see Materials and methods and Fig. 1B. (B) The latency difference between stimulation and non-stimulation trials for the generation of visually guided saccades to the target is plotted as a function of target diameter for three levels of current: 25, 50, and 100 μA. Data for the 100-μA current level are from A. See A for other details.

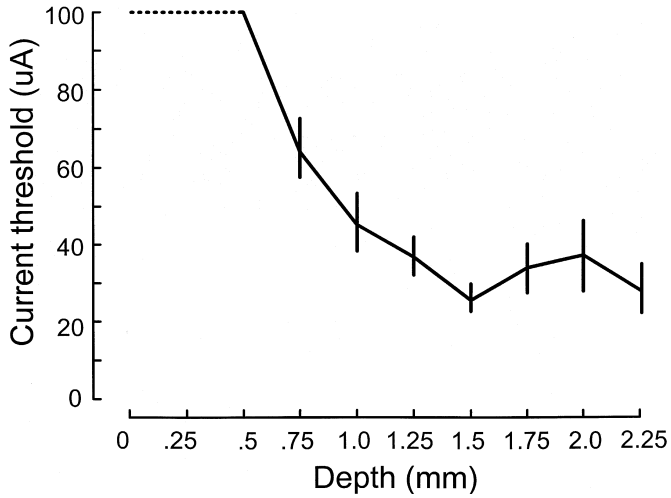

FIG. 7. Current threshold for inducing a delay in the execution of visually guided saccades to a target in the receptive field of stimulated neurons. The current threshold for inducing a 20-ms increase in latency for saccades generated to a target located at the receptive-field location of the stimulated neurons is plotted as a function of cortical depth for seven penetrations made into V1 of monkey M and three penetrations made into monkey D. The solid curve represents the average threshold value and the dotted portion of the curve indicates that a 100 μA current was not sufficient for inducing a 20-ms latency increase. Standard errors are shown. To derive a threshold value for a given site, blocks of 20 stimulation trials were interleaved with 20 non-stimulation trials using currents above and below threshold. The 20-ms criterion is based on data shown in Fig. 4C. Here the average latency difference of 20 ms is greater than three standard deviations of that observed when comparing non-stimulation and dummy stimulation trials (standard deviation = 4.3 ms, $n = 35$). For all stimulation trials, anode-first pulses were used and the train duration, pulse duration, and pulse frequency were fixed at 100 ms, 0.2 ms, and 200 Hz, respectively. To determine the current threshold, the following current levels were used: 2, 5, 10, 15, 25, 50, 75, and 100 μA. The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. See Materials and methods and Fig. 1B for other details.

Discussion

The central finding we report here is that when electrical stimulation is delivered to V1 during active fixation, the execution of a saccadic eye movement made to a visual target positioned in the receptive field of the stimulated neurons is delayed. No delay is obtained when the visual target is placed at locations outside of the receptive field. The optimal parameters of stimulation to induce this delay are as follows: (i) anode-first pulses are more effective than cathode-first pulses; (ii) train durations greater than 40 ms and frequencies greater than 100 Hz are necessary to get the best effect and (iii) stimulation has to be delivered at the end of the fixation period, immediately before the presentation of the visual target. Also the most pronounced saccadic delay occurs when the visual target is centred on the receptive-field location of the stimulated neuron, when the target is less than 0.4 degrees in diameter, and when the stimulating electrode is positioned at 1.5 mm below the cortical surface (i.e. the depth at which the current threshold to induce a 20-ms saccadic delay was lowest).

Volume of tissue activated to evoke the saccadic delay

Based on the data of Fig. 5C, we now have an estimate of how far current of 25, 50, and 100 μA spread by noting the amount of the visual field that is affected by these currents at an eccentricity of four degrees (i.e. the centre of the receptive-field in this case). From each curve in the figure, we determined the amount of visual field affected by measuring the target eccentricity affected at 50% of the maximal latency difference. For the 25, 50, and 100 μA currents, the amount of visual field affected by the stimulation was 0.31, 0.39, and 0.58 degrees, respectively. At a four-degree eccentricity, 0.5 degrees of the visual field span two ocular dominance columns (LeVay *et al.*, 1985), which represent 800 μm of V1 tissue (Wiesel *et al.*, 1974; LeVay *et al.*, 1975, 1985; Blasdal & Salama, 1986). Therefore, 25, 50, and 100 μA affects V1 tissue within 248, 312, and 464 μm from the electrode tip. These estimates are somewhat greater than estimates obtained by using the current–distance equation [25, 50, and 100 μA currents affect V1 tissue within 158, 224, and 316 μm from the electrode tip (see Results)]. This difference is likely related to the fact that the current–distance equation is based on the delivery of a single pulse, whereas the experiments described here are based on the delivery of 20 pulses using a 100 ms pulse train. For details about transynaptic effects due to multiple pulses in a train see McIlwain (1982).

Excitability properties of elements mediating the saccadic delay

One way of measuring the excitability of stimulated elements is to determine their chronaxie, which is an estimate of the time constant of the directly stimulated neural elements (Ranck, 1975). Chronaxies have been determined for elements mediating fMRI signal, neurotransmitter release, classical conditioning, self-stimulation, phosphene induction, and saccadic eye movements (Brindley & Lewin, 1968a; Dobelle & Mladejovsky, 1974; Matthews, 1977; Deyoe, 1983; Tehovnik & Lee, 1993; Farber *et al.*, 1997; Tehovnik & Sommer, 1997; Tehovnik *et al.*, 2003a; Tolias *et al.*, 2003).

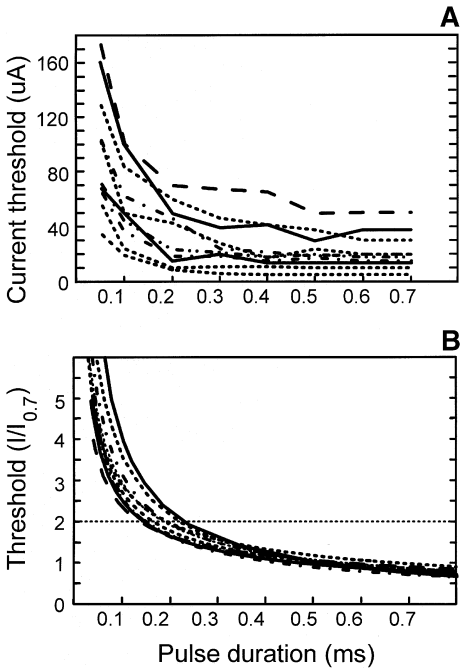

FIG. 8. The excitability of neurons mediating the delay in visually guided saccades to a target located in the receptive field of the stimulated neurons. (A) The current threshold for inducing a 20-ms increase in latency for saccades generated to a target located at the receptive-field location of the stimulated neurons is plotted as a function of pulse duration. Each curve represents data from a different stimulation site: six sites from monkey M and four sites from monkey D. Sites were located from 0.5 to 1.5 mm below the cortical surface. To derive a point on a curve, blocks of 20 stimulation trials interleaved with 20 non-stimulation trials were conducted using currents above and below threshold. The 20-ms criterion is based on data shown in Fig. 4C. Here the average latency difference of 20 ms is greater than three standard deviations of that observed when comparing non-stimulation and dummy stimulation trials (standard deviation = 4.3, $n = 35$). The target used was brighter than background at 100% contrast and was 0.2 degrees in diameter. See Material and methods and Fig. 1B for other details. (B) Normalized threshold current based on the data from above is plotted as a function of pulse duration using power functions. For a pulse duration of 0.7 ms, the current required to induce a 20-ms latency shift is set to one and all other values are expressed as a multiple of the current used at the 0.7-ms pulse duration. The pulse duration at which a curve intersects two units of threshold (designated by the dotted horizontal line) indicates the chronaxie of the stimulated elements at a site of study.

The chronaxie values of the directly stimulated elements mediating the saccadic delay in V1 ranged from 0.13 to 0.24 ms. This range of values overlaps with that reported for cortical pyramidal neurons which range from 0.1 to 0.4 ms (Stoney *et al.*, 1968; Asanuma *et al.*, 1976). Also the range of values overlaps with that reported for elements mediating stimulation-evoked saccades induced from V1 (0.08–0.41 ms; Tehovnik *et al.*, 2003a). This suggests that the directly stimulated elements mediating the saccadic delay are pyramidal and similar to those that mediate saccades evoked from V1.

Another method of assessing neuronal excitability is to determine whether the polarity of pulses differentially affects the chances of evoking a particular response. We found that anode-first pulses were superior to cathode-first pulses for inducing a saccadic delay. This suggests that the directly stimulated elements are comprised mainly of cell bodies or terminals over axons (Porter, 1963; Stoney *et al.*, 1968; Armstrong *et al.*, 1973; Clendenin *et al.*, 1974; Ranck, 1975; Rattay, 1999; McIntyre & Grill, 2000; Tehovnik & Slocum, 2003a; Tehovnik *et al.*, 2003a). Action potentials are induced by stimulation when an outward current is produced at the initial segment or nodes of Ranvier of a neuron. When an electrode is in a cell-body-rich or terminal-rich area an anodal pulse is most effective at inducting such an outward current, and when an electrode is in an axon-rich area a cathodal pulse is most effective at inducing such a current (Ranck, 1975).

Using electrical stimulation to interfere with visual processing

Electrical stimulation has been used effectively to determine regions of the brain that mediate target selection (Schiller & Tehovnik, 2001). These regions include areas V1 and V2, the lateral intraparietal area, and the medial and frontal eye fields. Two types of effects have been reported. First, the delivery of electrical stimulation was found to facilitate the selection of a visual target located in the receptive or motor field of the stimulated neurons when stimulation was delivered to the deep layers of V1 and V2, to various regions within the lateral intraparietal area, and to every site tested in the medial and frontal eye fields. Second, the delivery of stimulation was found to interfere with the selection of a visual target located in the receptive-field or motor-field location of the stimulated neurons when stimulation was delivered to the upper layers of V1 and V2, and to various sites within the lateral intraparietal area.

Also cortical inhibitory circuits have been shown to mediate target selection by presenting a visual target in the receptive or motor field of neurons injected with different GABAergic agents (Schiller & Tehovnik, 2003). The GABA agonist, muscimol, injected into V1 or the frontal eye fields produced interference. The GABA antagonist, bicuculline, injected into V1 caused interference, while such injections made into the frontal eye fields induced facilitation. Delivering either agent into the intraparietal cortex had little effect, but when high doses of muscimol were used there was a small interference effect. This is consistent with the observations of Wardak *et al.* (2002).

Further work in V1 revealed that the best sites for the induction of an interference effect occurred when stimulation was delivered to the intermediate layers of V1 (Tehovnik *et al.*, 2002a,b; Tehovnik & Slocum, 2003a,b). Delivering stimulation to this region not only affected target selection, but also impaired the detection of single targets located in the receptive field of the stimulated neurons. Accordingly, we have argued that the electrical stimulation of V1 was inducing an interference effect by disrupting visual processing in the geniculostriate system.

The idea that electrical stimulation of the geniculostriate pathway affects the flow of visual information is not new. The delivery of a single stimulation pulse to the optic tract or area V1 renders neurons in the lateral geniculate nucleus unresponsive to a visual stimulus for several tens of milliseconds (Schiller & Malpeli, 1977). The duration of the interference due to stimulation was much longer when the pulse was delivered to V1 as compared to when it was delivered to the optic tract. Also, the delivery of a single pulse to superficial V1 dampens mainly the responsivity of cells within V1, whereas stimulation of deeper regions of V1 affects cells in V1 as well as those in the lateral geniculate nucleus (Chung & Ferster, 1998). Chung & Ferster (1998) have suggested that the deeper stimulation might be activating the lateral geniculate nucleus antidromically from lamina IV as well as orthodromically via the corticothalamic projection originating from lamina VI.

Using a behavioural paradigm in which the electrical stimulation was delivered during active fixation (rather than concurrently with the presentation of the visual target), we found that the best interference (i.e. saccadic delay) occurred when pulses were delivered to the deepest layers of V1 (current study). By delivering electrical stimulation during active fixation, all electrically evoked

saccades are suppressed given that currents as high as 1500 μA fail to evoke saccades from V1 during active fixation (Tehovnik *et al*., 2003b). By inducing this suppression, the neuronal elements in the deepest layers of V1 that feedback onto the lateral geniculate nucleus might now be able to contribute to the interference effect. Indeed, the excitability properties of the elements that mediate the interference were similar to those of pyramidal neurons (current study). Stimulation of these elements may induce an interference effect by directly activating a combination of V1 circuits; pyramidal fibres intrinsic to V1 and pyramidal fibres from V1 that feedback onto the lateral geniculate nucleus. This idea is consistent with the observations of Schiller & Malpeli (1977) and Chung & Ferster (1998) and with known pyramidal projections between the striate cortex and the lateral geniculate nucleus (Lund *et al*., 1975). Whether the interference effect is due to both orthodromic and antidromic activation of the lateral geniculate nucleus needs to be ascertained electrophysiologically using established methods (e.g. Sommer & Wurtz, 2004).

Phosphene induction

From a previous set of experimental results (based on chronaxie estimates), we had suggested that the neuronal elements that mediate the execution of electrically evoked saccades from V1 in macaque monkeys are similar to those that mediate electrically evoked phosphenes in human V1 and that these elements exhibit a pyramidal morphology (Brindley & Lewin, 1968a; Dobelle & Mladejovsky, 1974; Tehovnik *et al*., 2003a). In the current study, we found that the chronaxie estimates of V1 elements that mediate the saccadic delay overlap with those that mediate electrically elicited saccades and additionally they overlap with those that mediate phosphenes in humans (i.e. having chronaxie values that fall between 0.1 and 0.4 ms). What this implies is that every time electrical stimulation is delivered to V1 in the macaque monkey either for the induction of a saccadic delay or for the production of a saccadic eye movement, a visual phosphene is produced by direct activation of pyramidal neurons. This concurs with the finding that simple and complex cells in V1 exhibit a pyramidal morphology (Gilbert & Wiesel, 1979) and that these cell types are readily activated by electrical stimulation (Ronner *et al*., 1981).

Electrical stimulation of V1 is ideal for encoding the location of a target in visual space. Currents of 100 μA to induce a saccadic delay are confined to one hypercolumn. With the use of microelectrodes, phosphenes can be evoked using currents much below 100 μA and even as low as 2 μA (Schmidt *et al*., 1996). The size of a phosphene increases systematically as stimulation is delivered to more peripheral representations of the V1 map (Brindley & Lewin, 1968a). The size of a phosphene is well within the range of sizes of receptive fields of V1 neurons in primates, i.e. all within three degrees of visual angle for the largest eccentricities (Dobelle & Mladejovsky, 1974; Hubel & Wiesel, 1977; Schmidt *et al*., 1996).

The challenge now will be to determine if electrical stimulation of V1 can selectively interfere with visual percepts that are uniquely coded by the directly stimulated neurons. Such effects should be best realized at the lowest current levels at which the fewest number of cells are excited.

Acknowledgements

We thank Ms. Christina Carvey for her help with the preparation of the manuscript. NIH EY08502 to P.H. Schiller supported this work.

Abbreviations

V1, primary visual cortex.

References

Armstrong, D.G., Harvey, R.J. & Schild, R.F. (1973) The spatial organization of climbing fibres branching in the cat cerebellum. *Exp. Brain Res.*, **18**, 40–58.

Asanuma, H., Arnold, A. & Zarzecki, P. (1976) Further study on the execution of pyramidal tract cells by intracortical microstimulation. *Exp. Brain Res.*, **26**, 443–461.

Blasdal, G.G. & Salama, G. (1986) Voltage-sensitive dyes reveal a modular organization in monkey striate cortex. *Nature*, **321**, 579–585.

Brindley, G.S. & Lewin, W.S. (1968a) The sensations produced by electrical stimulation of visual cortex. *J. Physiol. (Lond.)*, **196**, 479–473.

Brindley, G.S. & Lewin, W.S. (1968b) The visual sensations produced by electrical stimulation of the medial occipital cortex. *J. Physiol. (Lond.)*, **194**, 54P–55P.

Chung, S. & Ferster, D. (1998) Strength and orientation tuning of the thalamic input to simple cells revealed by electrically evoked cortical suppression. *Neuron*, **20**, 1177–1189.

Clendenin, M., Ekerot, C.-F., Oscarsson, O. & Rosen, I. (1974) The lateral reticular nucleus in the cat. I. Mossy fibre distribution in cerebellar cortex. *Exp. Brain Res.*, **21**, 473–486.

Cragg, B.G. (1967) The density of synapses and neurons in the motor and visual areas of the cerebral cortex. *J. Anat.*, **101**, 639–654.

Deyoe, E.A. (1983) An investigation in the awake macaque of the threshold for detection of electrical currents applied to striate cortex: psychophysical properties and laminar differences. PhD Thesis, University of Rochester, Rochester, N.Y.

Dobelle, W.H. & Mladejovsky, M.G. (1974) Phosphenes produced by electrical stimulation of humans occipital cortex, and their application to the development of a prosthesis for the blind. *J. Physiol. (Lond.)*, **243**, 553–576.

Farber, S.A., Bogdanov, M., Marshall, D.L. & Tehovnik, E.J. (1997) Excitability of neural elements within rat corpus striatum. *J. Neurosci. Meth.*, **76**, 93–104.

Finlay, B.L., Schiller, P.H. & Volman, S.F. (1976) Quantitative studies of single-cell properties in monkey striate cortex. IV. Corticotectal cells. *J. Neurophysiol.*, **39**, 1352–1361.

Fries, W. (1984) Cortical projections to the superior colliculus in the macaque monkey: a retrograde study using horseradish peroxidase. *J. Comp. Neurol.*, **230**, 55–76.

Gilbert, C.D. & Wiesel, T.N. (1979) Morphology and intracortical projections of functionally characterized neurons in the cat visual cortex. *Nature*, **280**, 120–125.

Hentall, I.D., Zorman, G., Kansky, S. & Fields, H.L. (1984) Relations among threshold, spike height, electrode distance, and conduction velocity in electrical stimulation of certain medullospinal neurons. *J. Neurophysiol.*, **51**, 968–977.

Hubel, D.H. & Wiesel, T.N. (1977) Functional architecture of macaque monkey visual cortex. *Proc. Royal Soc. Lond., B*, **198**, 1–59.

Jankowska, E. & Roberts, W.J. (1972) An electrophysiological demonstration of the axonal projections of single spinal interneurons in the cat. *J. Physiol. (Lond.)*, **222**, 597–622.

Judge, S.J., Richmond, B.L. & Chu, F.C. (1980) Implantation of magnetic search coils for measurement of eye position: an improved method. *Vis. Res.*, **20**, 535–538.

LeVay, S., Connolly, M., Houde, J. & van Essen, D.C. (1985) The complete pattern of ocular dominance stripes in the striate cortex and visual field of macaque monkey. *J. Neurosci.*, **5**, 486–501.

LeVay, S., Hubel, H. & Wiesel, T.N. (1975) The pattern of ocular dominance columns in macaque visual cortex revealed by reduced silver stain. *J. Comp. Neurol.*, **159**, 559–576.

Li, C.L. & Bak, A. (1976) Excitability characteristics of the A- and C-fibres in the peripheral nerve. *Exp. Neurol.*, **50**, 67–79.

Lund, J.S., Lund, R.D., Hendrickson, A.E., Bunt, A.H. & Fuchs, A.F. (1975) The origin of efferent pathways from the primary visual cortex, area 17, of the macaque monkey as shown by retrograde transport of horseradish peroxidase. *J. Comp. Neurol.*, **164**, 87–304.

Macpherson, J., Wiesendanger, M., Marangoz, C. & Miles, T.S. (1982) Corticospinal neurons of the supplementary motor area of monkeys: a single unit study. *Exp. Brain Res.*, **48**, 81–88.

Matthews, G. (1977) Neural substrate for brain stimulation reward in the rat: cathodal and anodal strength duration properties. *J. Comp. Physiol. Psych.*, **91**, 858–875.

McIlwain, J.T. (1982) Lateral spread of neural excitation during microstimulation in intermediate gray layers of cat's superior colliculus. *J. Neurophysiol.*, **47**, 167–178.

McIntyre, C.C. & Grill, W.M. (2000) Selective microstimulation of central nervous system neurons. *Ann. Biomed. Eng.*, **28**, 219–233.

Nowak, L.G. & Bullier, J. (1998) Axons, but not cell bodies, are activated by electrical stimulation in cortical grey matter. I. Evidence from chronaxie measurements. *Exp. Brain Res.*, **118**, 477–488.

O'Kusky, J. & Colonnier, M. (1982) A laminar analysis of the number of neurons, glia, and synapses in the visual cortex (area 17) of adult macaque monkeys. *J. Comp. Neurol.*, **210**, 278–290.

Peters, A. (1987) Number of neurons and synapses in primary visual cortex. In Jones, E.G. & Peters, A. (Eds), *Cerebral Cortex*, Vol. 6. Plenum Press, New York, pp. 267–294.

Porter, R. (1963) Focal stimulation of hypoglossal neurons in the cat. *J. Physiol. (Lond.)*, **169**, 630–640.

Ranck, J.B. Jr (1975) Which elements are excited in electrical stimulation of mammalian central nervous system: a review. *Brain Res.*, **98**, 417–440.

Rattay, F. (1999) The basic mechanism for the electrical stimulation of the nervous system. *Neuroscience*, **89**, 335–346.

Roberts, W. & Smith, D.O. (1973) Analysis of threshold currents during microstimulation of fibres in the spinal cord. *Acta Physiol. Scand.*, **89**, 384–394.

Rockel, A.J., Hiorns, R.W. & Powell, T.P.S. (1980) The basic uniformity in structure of the neocortex. *Brain*, **103**, 221–244.

Ronner, S.F., Foote, W.E. & Pollen, D.A. (1981) Intracortical microstimulation of neurons in the visual cortex of the cat. *Electroencephalogr. Clin. Neurophysiol.*, **52**, 375–377.

Schiller, P.H. & Malpeli, J.P. (1977) Shock-induced inhibition in the lateral geniculate nucleus of the rhesus monkey. *Brain Res.*, **137**, 387–389.

Schiller, P.H. & Tehovnik, E.J. (2001) Look and see: how the brain moves the eyes. *Prog. Brain Res.*, **134**, 127–142.

Schiller, P.H. & Tehovnik, E.J. (2003) Cortical inhibitory circuits in eye-movement generation. *Eur. J. Neurosci.*, **18**, 3127–3133.

Schmidt, E.M., Bak, M.J., Hambrecht, F.T., Kufta, C.V., O'Rourke, D.K. & Vallabhanath, P. (1996) Feasibility of a visual prosthesis for the blind based on intracortical microstimulation of the visual cortex. *Brain*, **119**, 507–522.

Sommer, M.A. & Wurtz, R.H. (2004) What the brainstem tells the frontal cortex. I. Oculomotor signals sent from superior colliculus to frontal eye field via mediodorsal thalamus. *J. Neurophysiol.*, **91**, 1381–1402.

Stoney, S.D., Thompson, W.D. & Asanuma, H. (1968) Excitation of pyramidal tract cells by intracortical stimulation: effective extent of stimulating current. *J. Neurophysiol.*, **31**, 659–669.

Tehovnik, E.J. (1996) Electrical stimulation of neural tissue to evoke behavioural responses. *J. Neurosci. Meth.*, **65**, 1–17.

Tehovnik, E.J. & Lee, K.-M. (1993) The dorsomedial frontal cortex of the rhesus monkey. Topographic representation of saccades evoked by electrical stimulation. *Exp. Brain Res.*, **96**, 430–442.

Tehovnik, E.J. & Slocum, W.M. (2003a) Microstimulation of macaque V1 disrupts target selection: effects of stimulation polarity. *Exp. Brain Res.*, **148**, 233–237.

Tehovnik, E.J. & Slocum, W.M. (2003b) Using ocular dominance to infer the depth of the visual input layers of V1 in behaving macaque monkey. *J. Neurosci. Meth.*, **125**, 121–128.

Tehovnik, E.J., Slocum, W.M. & Carvey, C.E. (2003b) Behavioural state affects saccadic eye movements evoked by microstimulation of striate cortex. *Eur. J. Neurosci.*, **18**, 969–979.

Tehovnik, E.J., Slocum, W.M. & Schiller, P.H. (2002a) Differential effects of laminar stimulation of V1 cortex on target selection by macaque monkeys. *Eur. J. Neurosci.*, **16**, 751–760.

Tehovnik, E.J., Slocum, W.M. & Schiller, P.H. (2002b) Microstimulation of V1 input laminae disrupts the selection and detection of visual targets. *Soc. Neurosci. Abstr.*, 55.4.

Tehovnik, E.J., Slocum, W.M. & Schiller, P.H. (2003a) Saccadic eye movements evoked by microstimulation of striate cortex. *Eur. J. Neurosci.*, **17**, 870–878.

Tehovnik, E.J. & Sommer, M.A. (1997) Electrically evoked saccades from the dorsomedial frontal cortex and frontal eye fields: a parametric evaluation reveals differences between areas. *Exp. Brain Res.*, **117**, 369–378.

Tolias, A.S., Augath, M., Pauls, J., Oeltermann, O., Tehovnik, E.J., Schiller, P.H. & Logothetis, N.K. (2003) Combined electrical microstimulation and fMRI in the macaque. *Soc. Neurosci. Abstr.*, 69.2.

Troyk, P., Bak, M., Berg, J., Bradley, D., Cogan, S., Erickson, R., Kufta, C., McCreery, D., Schmidt, E. & Towle, V. (2003) A model for intracortical visual prosthesis research. *Artificial Organs*, **27**, 1005–1015.

Wardak, C., Olivier, E. & Duhamel, J.R. (2002) Attentional deficit after lateral intraparietal area inactivation in monkey. *J. Neurosci.*, **22**, 9877–9884.

West, D.C. & Wolstencroft, J.H. (1983) Strength-duration characteristics of myelinated and non-myelinated bulbospinal axons in the cat spinal cortex. *J. Physiol. (Lond.)*, **337**, 37–50.

Wiesel, T.N., Hubel, D.H. & Lam, D.M.K. (1974) Autoradiographic demonstration of ocular dominance in the monkey striate cortex by means of transneuronal transport. *Brain Res.*, **79**, 273–279.